School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Project Scribe Report (Questions 1-3)

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Scribe Question 1: Project System and Objective
What is the probabilistic problem being addressed in your project? Clearly state the system objective and identify the primary sources of uncertainty involved.

Answer:

1 Project System and Objective

1.1 Probabilistic Problem

The distributed Denial of Service (DDoS) attacks are intended to inundate the victim with a large amount of messages.high number of packets of distributed and sometimes spoofed sources. Due to IP spoofing,partial visibility of the network paths, and packet sampling, the actual source of attack traffic.cannot determine how to be in deterministic form. Thus, in these situations, it is natural that IP traceback takes place. turns into a probabilistic and not deterministic inference problem.

1.2 Understand the Problem

SAMP packet, the Source Path Isolation Engine (SPIE) is used to gather information that is a sampler of the packet.packets whose probability to reduce storage and processing overhead is low (p = 3%). As aresult, partial and noisy observations of attack traffic only are available. Since not all packetsand paths are monitored, the traceback system will have to guess probable attack paths using this as a basis.

1.3 System objective

The Particle Swarm Optimization (PSO) is applied in the project to tackle this uncertainty.probabilistic IP traceback. In the proposed scheme, a candidate is represented by each particle.path of attack by the attacker to the victim. Depending on how, the fitness of a particle is determined.well the possible matching path corresponds to the packet information that is gathered by SPIE. Through PSO investigates iterative lessening relying on individual experience and the knowledge attained on swarm level. several potential paths and hastens down to the most likely attack routes.

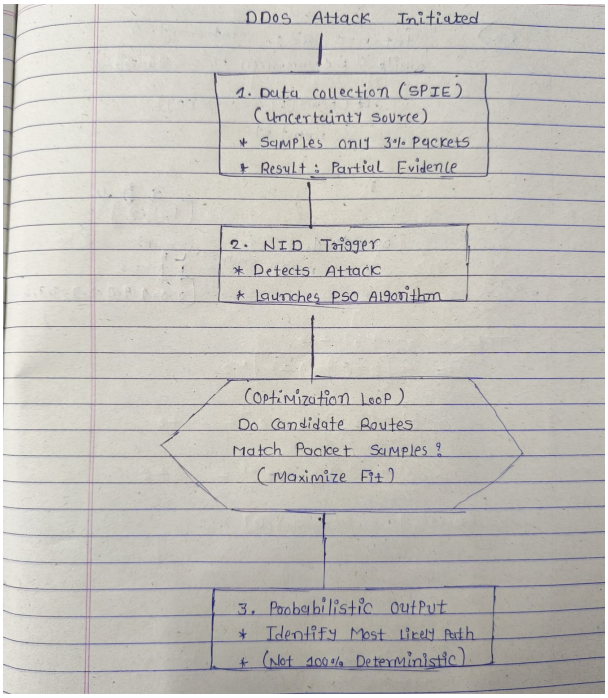

Figure 1: Caption

1.4 Primary Sources of Uncertainty

The primary aim of suggested PSO-based IP Traceback system (PSO-IPTBK) is to effectively narrow down on the most probable source tracks of a DDoS attack using incomplete and information about packets is insecure. The system has less exhaustive traceback compared to the methods. seeks to provide computational efficiency and reasonable accuracy in the real.

Scribe Question 2: Key Random Variables and Uncertainty Modeling
Identify the key random variables in your project and describe how uncertainty is modeled for each. Clearly state any probabilistic assumptions currently being made.

Answer:

2 Key Random Variables and Uncertainty Modeling

2.1 Key Random Variables

We have modeled the system using two primary sets of random variables:

- **The Packet Sampling Variable (S):** Since we are using the Source Path Isolation Engine (SPIE) mechanism, the presence of a packet in our traceback log is a random variable modeled as a Bernoulli trial with a success probability of $p = 3\%$. This means our dataset is inherently incomplete.
- **The Stochastic Coefficients (Randomness in Motion):** The core random variables in our system are the `rand()` functions embedded in the Particle Swarm Optimization (PSO) velocity update equation. As defined in Equation (1) of the paper, the movement of a candidate route (V_i) is calculated as:

$$v_i(t+1) = w_i \cdot v_i(t) + c_1 \cdot \text{rand}() \cdot (P_{best} - x_i(t)) + c_2 \cdot \text{rand}() \cdot (P_{gbest} - x_i(t)) \tag{1}$$

2.2 Modeling Uncertainty

We model uncertainty as an Optimization Problem. Since the True Attack Path is unknown due to IP Spoofing, we cannot calculate it deterministically. Instead, we model the likelihood of a path using a Fitness Function (F).

The uncertainty is reduced by minimizing the Euclidean distance between a particle's current position and the swarm's global best position. We utilize Equation (5) from the paper to measure this convergence:

$$D(x_i, P_{\text{gbest}})_j = \sqrt{\sum_{i=1}^{n} [x_i(i,j) - P_{\text{gbest}}(i,j)]^2} \tag{2}$$

If this distance decreases over time ($D_{t+1} < D_t$), the uncertainty regarding that specific route is reduced, and the fitness value increases.

Scribe Question 3: Probabilistic Reasoning and Dependencies
Explain how probabilistic relationships (e.g., dependence, independence, conditionality) are used within your project system to support reasoning, inference, or decision-making.

Answer:

3 Probabilistic Reasoning and Dependencies

3.1 Probabilistic Reasoning

DDoS attack case, the network behavior is extremely unpredictable as it is caused by IP spoofing, packet.Dynamic routing parameters, sampling, and dynamic routing conditions. The victim sees partial and noisy only, given this.there is no deterministic identification of the attack route and it is so because of packet information. Therefore the offered system is based on probabilistic arguments to deduce the most probable attack patterns.but, rather than trying to fix upon one definite path.The IP traceback system based on the PSO presents a traceback system as an optimization problem under.uncertainty. This model presumes an object that is attacked in every possible route as a particle.records of packets gathered by SPIE. The fitness of a particle is an indicator of the fitness.the observed packets support the corresponding route. Pathways that are finding direction to a.higher number of observations of packet is given more likelihood and is spared in preference during.the optimization process.

3.2 Depandencies

Dependencies are significant to the PSO algorithm. Each particle's movementis dependent on its personal best position (Pbest) which is the best route in the past.discovered by that particle according to observation of the past. This gives rise to conditional dependence.on the own search history of the particle. Moreover, the global best affects the particles.Photo (Pgbest), which is the most likely path discovered by the complete swarm,achieving a swarm dependency between particles. Undeterministic elements of update.In equations reflected stochasticity is controlled, and the system is free to be experimented.paths and escape an early meeting.

3.3 Conditionality

The packets are reversed through the network as traced back in the system model.The PSO method fits the candidate routes with the purpose of optimization. If the fitness values do not improve with repeated repetition, it shows that there is an attendance at a local optimum. The randomness and swarm interaction increase are used to get the system out of these local optima.and keep tried additional likely avenues of attack.All in all, probabilistic reasoning allows the system to deal with incomplete successfully.PSO has dependency mechanisms that guide the search and information and uncertainty.in a dynamically changing network setting, towards the most probable DDoS attack paths.

End of Scribe Report Submission

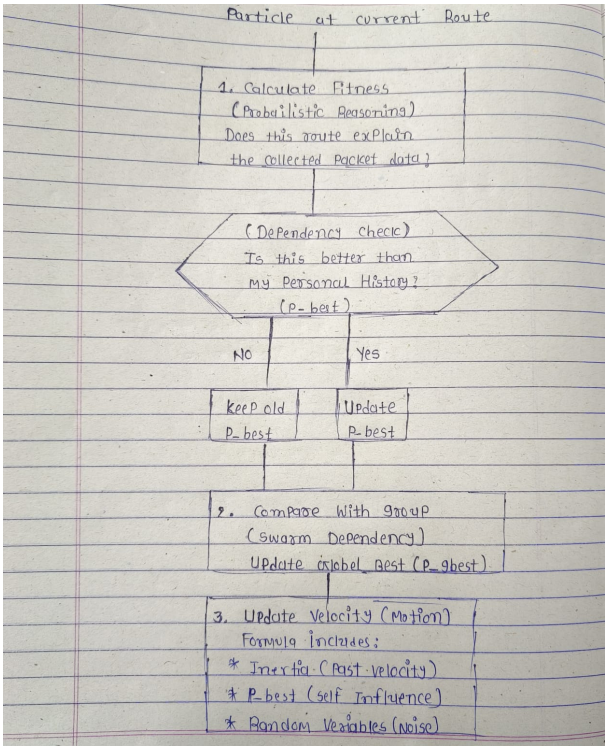

Figure 2: Caption